

CEL: a controlled benchmark for counterfactual explanations

Compare CE methods under one protocol across validity, proximity, plausibility, sparsity, and runtime.

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Code & project

BENCHMARK SCOPE

One broad benchmark

18
datasets

14
methods

3
paradigms

2
backbones / task

5
folds

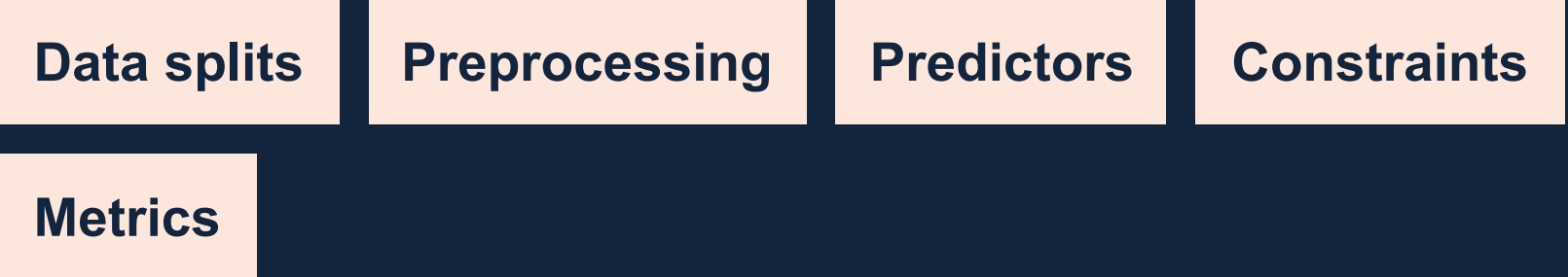
Source · Datasets · Methods · Full-Results

LEFT

Why CE results are hard to compare

Protocol variation can make setup effects look like method effects.

Are we measuring the CE method - or its setup?



Source · Benchmark

BOTTOM

Three contributions

Controlled evaluation. Broad CE coverage. An extensible open-source benchmark workbench.

CONTROL CEL standardizes datasets, preprocessing, predictive backbones, constraints, and metric definitions for fairer CE comparison.

COVER CEL benchmarks local, global, and group-wise counterfactual explanations inside one framework.

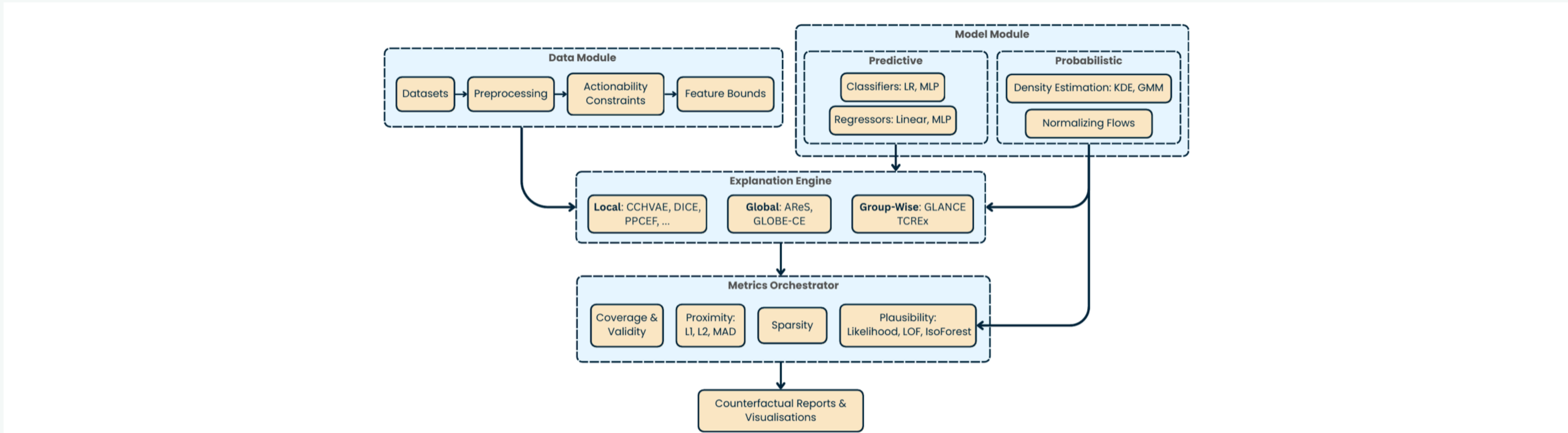
EXTEND The CEL library supports method integration, transparent reporting, and community-driven benchmarking.

Source · Introduction

CENTER

A benchmark, not just a library

The manuscript architecture joins data, models, three CE paradigms, and shared metrics.



MANUSCRIPT ARCHITECTURE

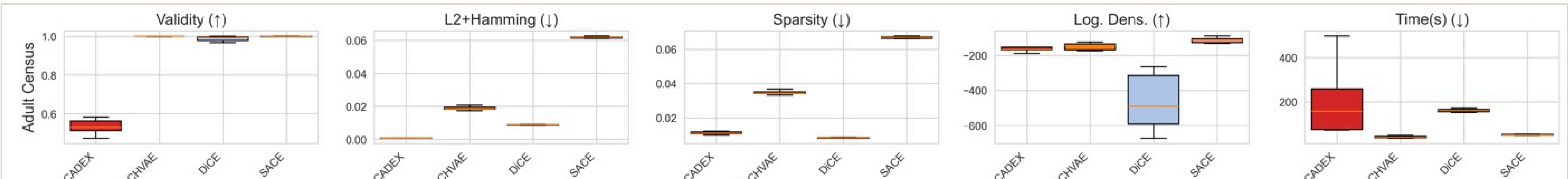
Source · Introduction

Data, models, CE methods, and metrics share one evaluation system.

RIGHT

Local methods: quality is multi-dimensional

High validity does not imply the smallest change, strongest density, or lowest runtime.



MANUSCRIPT LOCAL-METHOD FIGURE - ADULT CENSUS ROW

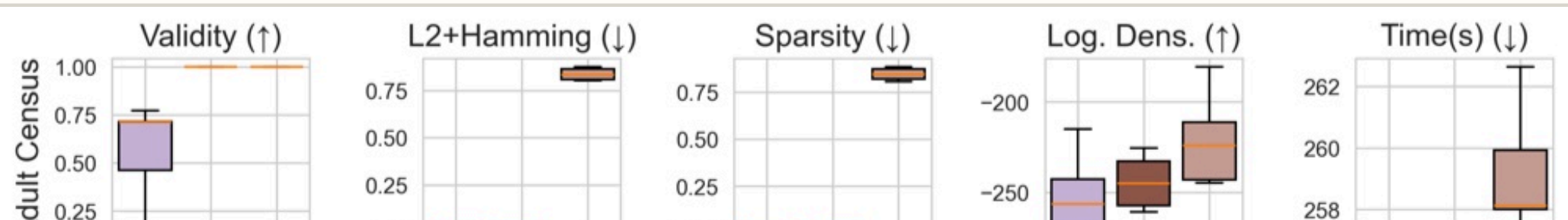
Resolution: omit any poster sparsity ranking; retain the manuscript figure without adding a comparative sparsity claim.

Source · Results · Conclusions · Metrics

BOTTOM

Global methods expose failure modes

GLOBE-CE and GlobalGLANCE show stronger validity; AReS is less reliable and frequently has missing outputs.



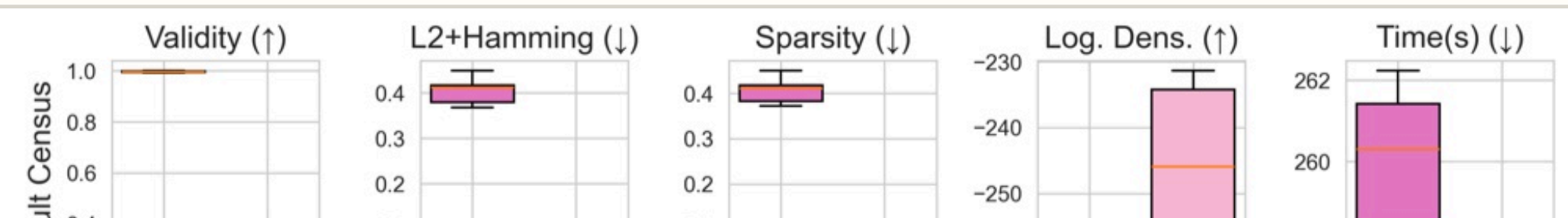
MANUSCRIPT GLOBAL-METHOD FIGURE - ADULT CENSUS ROW

Source · Results · tab:global_metrics_mlp

RIGHT

Group-wise: effectiveness versus minimal change

GLANCE reaches more instances; T-CREx favors smaller, more plausible shifts when it applies.



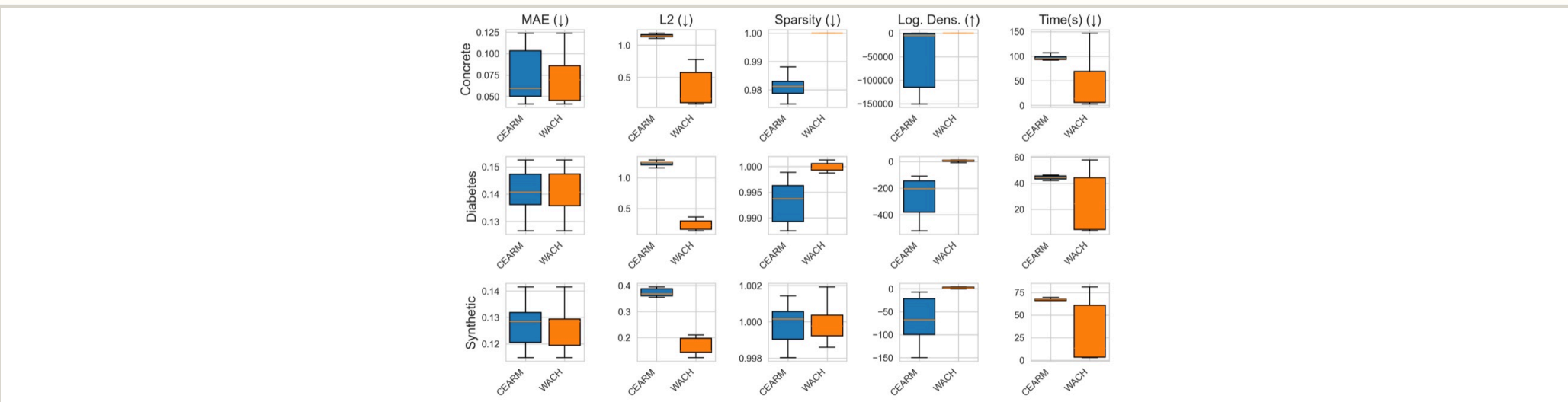
MANUSCRIPT GROUP-WISE FIGURE - ADULT CENSUS ROW

Source · Results · tab:group_metrics_mlp

BOTTOM

Regression: target accuracy is not enough

Wachter pairs comparable target error with smaller changes, stronger plausibility, and lower runtime in the manuscript comparison.



MANUSCRIPT REGRESSION-METHOD FIGURE

Source · Results · tab:regression_all_dnn